

Some trinomials are perfect squares. When they are factored, the two binomial factors are identical. These trinomials are called **trinomial squares**.

$$(x+5)(x+5) = (x+5)$$

Completing the square is a technique for solving quadratic equations that uses trinomial squares. Study the example to see how to make a trinomial square.

Hide
$$x^2 + 8x + \underline{\hspace{2cm}}$$

Take the number from the middle term: 8

Divide by 2.

$$8 \div 2 = 4$$

Square the result.

$$4^2 = 16$$

Fill in the blank.

$$x^2 + 8x + 16$$

Now you are ready to solve an equation by completing the square.

Hide
$$x^2 - 12x + 20 = 7$$

Make a space to complete the square.

Subtract 20 from both sides of the equation.

$$x^2 - 12x + 20 = 7$$

$$x^2 - 12x = -13$$

What number completes the perfect square?

Divide the number of the middle term by 2. Square the result.

$$-12 \div 2 = -6$$

$$(-6)^2 = 36$$

Fill in the blank. Add 36 to both sides.

$$x^2 - 12x = -13$$

$$x^2 - 12x + 36 = 23$$

Factor the left side.

$$(x - 6)^2 = 23$$

Take the square root of both sides.

$$\sqrt{(x - 6)^2} = \sqrt{23}$$

$$x - 6 = \pm \sqrt{23}$$

Isolate the variable. Add 6 to both sides.

$$x = 6 \pm \sqrt{23}$$

The symbol $\pm$ means “plus or minus.”

There are two solutions: $6 + \sqrt{23}$ and $6 - \sqrt{23}$.

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You can use your calculator to find the value of the solutions. Since the solutions represent irrational numbers, you will need to round to a convenient decimal place.

$$6 + \sqrt{23} \approx 10.7958$$

$$6 - \sqrt{23} \approx 1.2042$$